

Aviation Incident Analysis Engine

Select Analysis Mode: 

-  Diagnosis Mode
-  Prognosis Mode

Enter a description of an aviation incident or observation below. The system will analyze it against a database of historical NTSB reports to find related incidents and

Diagnosis Method

Choose diagnosis approach: 

- Similarity-Weighted (Current)
- Conditional Probability with Chain Rule (Professor's Method)

Enter Incident Description:

engine fire during takeoff

Diagnose Incident

Diagnostic Results

How Diagnosis Works (Overview): Your query is converted into an embedding (1,536-dimensional vector) using OpenAI's text-embedding-3-small model, then compared to other incident embeddings using cosine similarity. The top 50 most similar incidents are retrieved and grouped into pre-computed clusters (K-means, 50 clusters). Within each cluster, the most common root causes are identified. Finally, we combine all clusters using the chain rule: $P(\text{Cause}|\text{Query}) = \sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query})$.

Incident Clustering Analysis

Clustered 48 incidents into 6 types

Cluster Type	P(Cluster Query)	Incidents	Avg Similarity
Engine Fire	53.6%	26	0.631
Turbine Blade Failure	27.6%	13	0.651
Evacuation Slide Malfunction	6.3%	3	0.640
Loss Of Control	4.3%	2	0.657
Ice Ingestion	4.1%	2	0.632
Insufficient Information	0.0%	2	0.638

 View Detailed Cluster Breakdown

How these percentages are calculated: For each cluster, we collect all cause entries from every incident in that cluster. The percentage shown is $P(\text{Cause}|\text{Cluster}) = (\text{number of times the cause appears in the cluster}) / (\text{total number of cause entries in the cluster})$. For example, if a cluster has 33 total cause entries and a cause appears twice, its probability is $2/33 = 6.1\%$. Causes that appear only once will share the same percentage as other single-occurrence causes.

engine fire (26 incidents, ~23 total cause entries)

Top causes in this cluster:

- 4.3% (1 of 23 entries): LEAKAGE OF THE NUMBER 7 FUEL NOZZLE, WHICH RESULTED IN AN IN FLIGHT ENGINE FIRE
 - 4.3% (1 of 23 entries): THE TURBINE GUIDE VANE FAILURE
 - 4.3% (1 of 23 entries): THE MALFUNCTION OF THE STARTER CONTROL VALVE
-

turbine blade failure (13 incidents, ~34 total cause entries)

Top causes in this cluster:

- 8.8% (3 of 34 entries): Personnel issues-Action/decision-Action-Incorrect action performance-Maintenance...
 - 8.8% (3 of 34 entries): Personnel issues-Task performance-Maintenance-Installation-Maintenance personnel...
 - 8.8% (3 of 34 entries): The fracture of the fan mid shaft due to a stress corrosion cracking initiated b...
-

evacuation slide malfunction (3 incidents, ~2 total cause entries)

Top causes in this cluster:

- 50.0% (1 of 2 entries): A passenger sustained a hairline fracture during an emergency evacuation of the ...
 - 50.0% (1 of 2 entries): THE HIGH CYCLE FATIGUE FAILURE OF BLADES ON THE STAGE 1 HIGH PRESSURE COMPRESSOR...
-

loss of control (2 incidents, ~9 total cause entries)

Top causes in this cluster:

- 11.1% (1 of 9 entries): The HPT stage 2 disk failed because of low-cycle fatigue cracks that initiated f...
 - 11.1% (1 of 9 entries): Organizational issues-Development-Manufacture/production-Equipment manufacture-M...
 - 11.1% (1 of 9 entries): Aircraft-Aircraft power plant-Engine (turbine/turboprop)-Turbine section-Failure...
-

insufficient information (2 incidents, ~0 total cause entries)**ice ingestion** (2 incidents, ~4 total cause entries)

Top causes in this cluster:

- 25.0% (1 of 4 entries): Aircraft-Aircraft power plant-Engine fuel and control-Fuel controlling system-Ma...
 - 25.0% (1 of 4 entries): There was a malfunction of the electronic engine control (EEC) burner pressure (...)
 - 25.0% (1 of 4 entries): Aircraft-Aircraft power plant-Engine fuel and control-Fuel control electronic-Ma...
-



Root Cause Diagnosis

Method: Conditional probability with chain rule: $P(Cause|Query) = \sum P(Cause|Cluster) \times P(Cluster|Query)$

Analyzed 48 similar historical NTSB incidents

Step 1: Analyze Causes within Clusters

What's happening here: For each cluster, we look at the 'findings/causes' field from all incidents in that cluster. We count how often each cause appears and divide by the total number of incidents in that cluster to get $P(Cause|Cluster)$. This tells us: given this type of incident, what are the most likely root causes?

For each cluster, we identify the conditional probability of causes: $P(Cause|Cluster)$

 Engine Fire (P=53.6%)

Conditional Probabilities for Engine Fire:

- LEAKAGE OF THE NUMBER 7 FUEL NOZZLE, WHICH RESULTED IN AN IN FLIGHT ENGINE FIRE 4
- THE TURBINE GUIDE VANE FAILURE 4
- THE MALFUNCTION OF THE STARTER CONTROL VALVE 4
- Contributing to the failure was a combination of manufacturing defects of the fuel nozzle adapter as 4
- Contributing to the incident was a second defect of the fuel nozzle adapter assembly by the manufact 4
- Aircraft-Aircraft power plant-Engine fuel and control-Fuel distribution-Failure - C 4
- The sealing failure between the fuel nozzle adapter assembly and fuel transfer tubes that allowed fuel to leak into the nacelle fire zone where it was subsequently ignited by the hot combustor case. Contributing to the failure was a combination of manufacturing defects of the fuel nozzle adapter assembly by the manufacturer and incorrectly overhauled fuel transfer tubes by the engine manufacturer. Contributing to the incident was a second defect of the fuel nozzle adapter assembly by the manufacturer which prevented maintenance personnel from detecting the internal fuel leak. 4
- The loss of power in the No 4
- The loss of power in the No. 2 engine for undetermined reasons, and the inability of the pilot to establish a climb and/or maintain altitude. A factor was the incompletely feathered No. 2 propeller.. 4

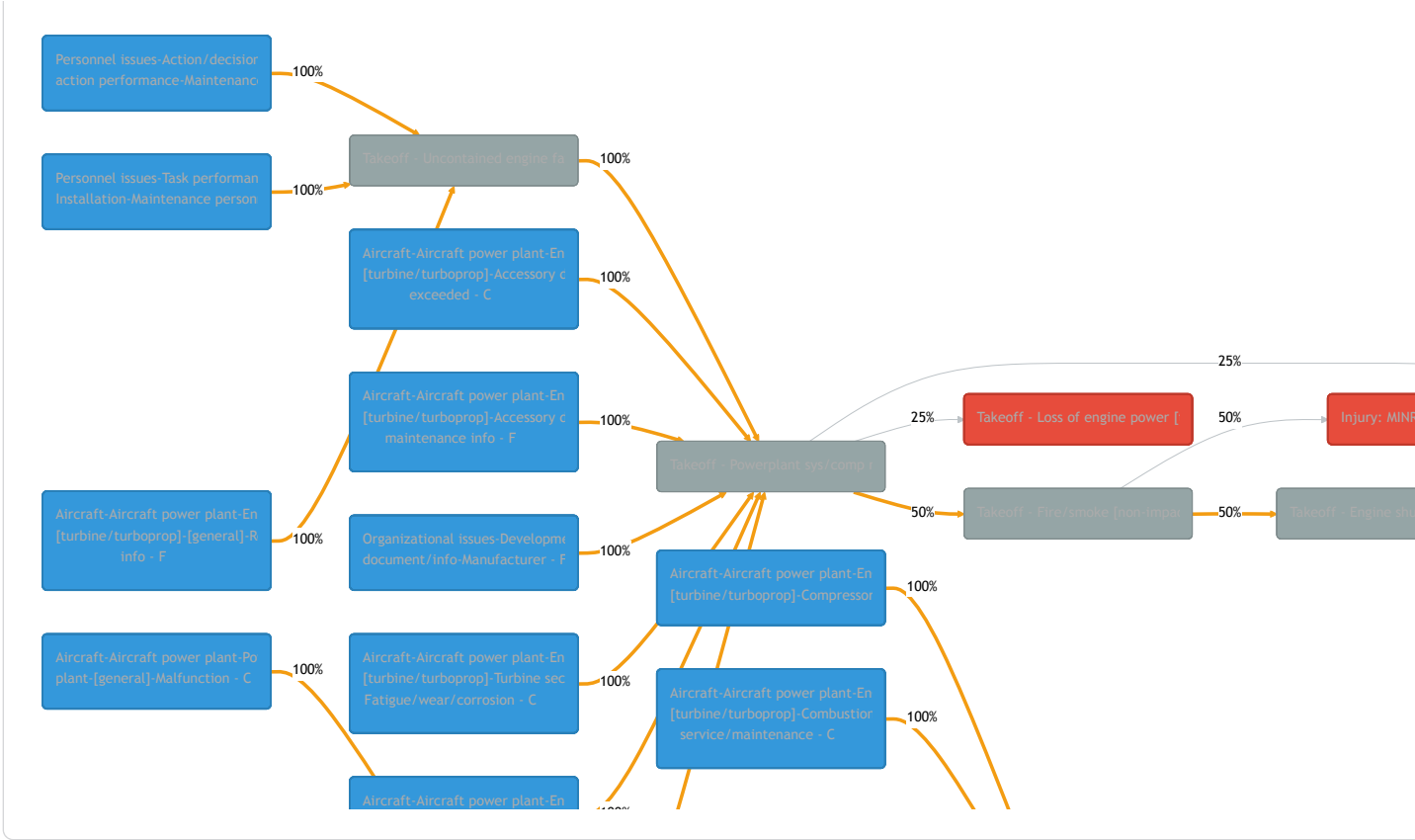
- a loose fuel nozzle 4
- Turbine Blade Failure (P=27.6%)

Conditional Probabilities for Turbine Blade Failure:

- Personnel issues-Action/decision-Action-Incorrect action performance-Maintenance personnel - C 8
- Personnel issues-Installation-Maintenance personnel - C 8
- The fracture of the fan mid shaft due to a stress corrosion cracking initiated by local regions of corrosion pitting, which resulted from oil migration within the shaft and its subsequent degradation. 8
- [turbine/turboprop]-[general]-Ir 5
- 2 engine failure and subsequent aborted takeoff as the high cycle fatigue failure of a fan blade ne 5
- The probable cause of the No. 2 engine failure and subsequent aborted takeoff was the high cycle fatigue failure of a fan blade near the blade root that resulted in extensive low pressure compressor damage and compressor case breach. 5
- A contributing factor was the inadequate diffuser and HPT case inspection procedure 5
- Aircraft-Aircraft power plant-Engine (turbine/turboprop)-Combustion section-Incorrect service/maintenance - C 5
- the return to service of overhauled airplane engine parts with undetected fatigue cracks, resulting in complete loss of engine power and uncontained engine failure. A contributing factor was the inadequate diffuser and HPT case inspection procedure. 5
- Aircraft-Aircraft power plant-Engine (turbine/turboprop)-Accessory drives-Capability exceeded - C 5
- Fuel leaking from the fuel flow transmitter that ignited on contact with the engine's hot cases which resulted in an engine undercowl fire. The fuel leak resulted from the fuel flow transmitter end housing backing off from the main housing, creating a gap that allowed high pressure fuel to escape. Under certain assembly variances and operational load and temperature variations, the end housing joint may not be able to maintain the preload without eventually yielding, which results in the loosening of the joint and the resultant gap. 5

Analysis of Precursors & Event Sequence:

Precursors & Causal Chain:



Evacuation Slide Malfunction (P=6.3%)

Conditional Probabilities for Evacuation Slide Malfunction:

- A passenger sustained a hairline fracture during an emergency evacuation of the airplane 5
- THE HIGH CYCLE FATIGUE FAILURE OF BLADES ON THE STAGE 1 HIGH PRESSURE COMPRESSOR, WHICH RESULTED IN BLADE FRACTURE AND SUBSEQUENT INTERNAL ENGINE DAMAGE AND FIRE. 5

Analysis of Precursors & Event Sequence:

Loss Of Control (P=4.3%)

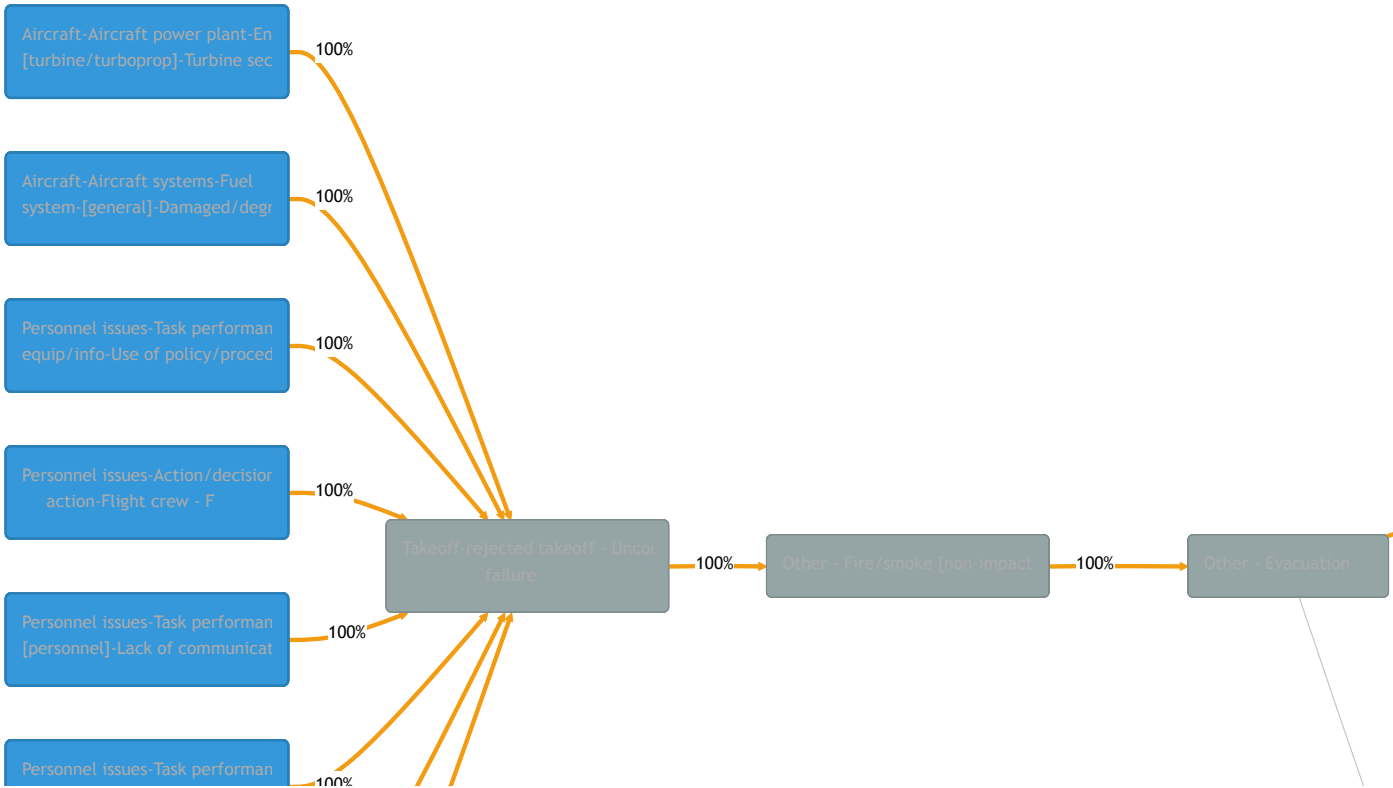
Conditional Probabilities for Loss Of Control:

- The HPT stage 2 disk failed because of low-cycle fatigue cracks that initiated from an internal subs 1
- Organizational issues-Development-Manufacture/production-Equipment manufacture-Manufacturer - C 1
- Contributing to the delay in shutting down the left engine was (1) the lack of a separate checklist 1
- The failure of the high-pressure turbine (HPT) stage 2 disk, which severed the main engine fuel feed line and breached the right main wing fuel tank, releasing fuel that resulted in a fire on the right side of the airplane during the takeoff roll. The HPT stage 2 disk failed because of low-cycle fatigue cracks that initiated from an internal subsurface manufacturing anomaly that was most likely not detectable during production inspections and subsequent in service inspections using the procedures in place. Contributing to the serious passenger injury was (1) the delay in shutting down the left engine and (2) a flight attendant's deviation from company procedures, which resulted in passengers evacuating from the left overwing exit while the left engine was still operating. Contributing to the delay in shutting down the left engine was (1) the lack of a separate checklist procedure for 1

- Boeing 767 airplanes that specifically addressed engine fires on the ground and (2) the lack of communication between the flight and cabin crews after the airplane came to a stop.
- Contributing to this accident was the Federal Aviation Administration’s failure to require the insta 1
 - Aircraft-Aircraft systems-Oxygen system-Passenger oxygen system-Design - C 1
 - The lack of positive separation allowed a short circuit to breach a combustibile oxygen hose, release 1
 - The design of the supplemental oxygen system hoses and the lack of positive separation between electrical wiring and electrically conductive oxygen 1
system components. The lack of positive separation allowed a short circuit to breach a combustibile oxygen hose, release oxygen, and initiate a fire in the supernumerary compartment that rapidly spread to other areas. Contributing to this accident was the Federal Aviation Administration’s failure to require the installation of nonconductive oxygen hoses after the safety issue concerning conductive hoses was initially identified by Boeing.

Analysis of Precursors & Event Sequence:

 Precursors & Causal Chain:



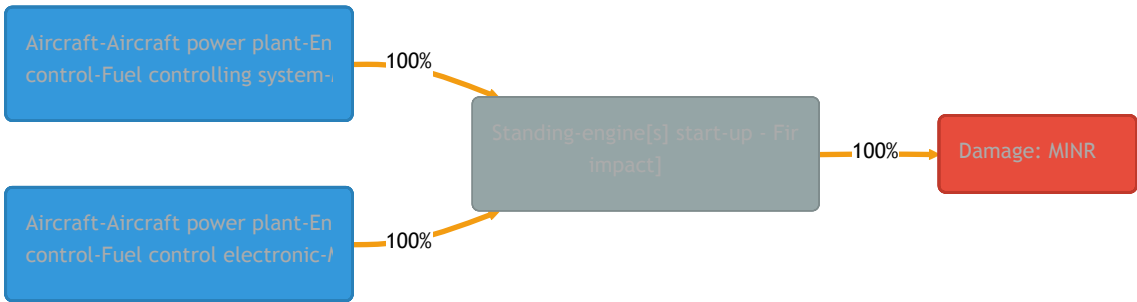
 Ice Ingestion (P=4.1%)

Conditional Probabilities for Ice Ingestion:

- Aircraft-Aircraft power plant-Engine fuel and control-Fuel controlling system-Malfunction - C 2
- There was a malfunction of the electronic engine control (EEC) burner pressure (Pb) sensor that had an abnormally high indication that resulted in an excessive amount of fuel being pumped into the engine that was still at a sub-idle condition. Because the engine was still accelerating to idle, the limited airflow through the engine engine was insufficient for the amount of fuel and resulted in a tailpipe fire. The cause of the EEC's Pb sensor fault could not be determined. 2
- Aircraft-Aircraft power plant-Engine fuel and control-Fuel control electronic-Malfunction - C 2
- the failure of the right engine's fuel control unit to operate correctly when the fuel metering valve became frozen in the full open position because of corrosion on the fuel metering valve slide and sleeve, which caused the engine to be overfueled during engine start and resulted in a tailpipe fire. 2

Analysis of Precursors & Event Sequence:

Precursors & Causal Chain:



Step 2: Final Root Cause Diagnosis

What's happening here: We combine results from ALL clusters using the chain rule (Law of Total Probability). For each possible cause, we multiply P(Cause|Clus cluster), then sum them up. This gives one final probability that weighs each cluster's contribution by how relevant it is to the query. Click on any cause below to

Combined probability using the Total Probability Theorem: $P(C|Q) = \sum P(C|I) \times P(I|Q)$

1. A passenger sustained a hairline fracture during an emergency evacuation of the airplane

Calculation: $\sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.5000 \times 0.0627) = 0.0313 \text{ (3.1\%)}$

View cluster-by-cluster breakdown

Evacuation Slide Malfunction: $P(C|\text{Cluster})=0.5000 \times P(\text{Cluster}|Q)=0.0627 = 0.0313$

2. THE HIGH CYCLE FATIGUE FAILURE OF BLADES ON THE STAGE 1 HIGH PRESSURE COMPRESSOR, WHICH RESULTED IN BLADE FRACTURE AND SUBSEQUENT INTERNAL ENGINE DAMAGE AND FIRE.

Calculation: $\sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.5000 \times 0.0627) = 0.0313 \text{ (3.1\%)}$

View cluster-by-cluster breakdown

Evacuation Slide Malfunction: $P(C|\text{Cluster})=0.5000 \times P(\text{Cluster}|Q)=0.0627 = 0.0313$

3. Personnel issues-Action/decision-Action-Incorrect action performance-Maintenance personnel - C

Calculation: $\sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0882 \times 0.2760) = 0.0244 \text{ (2.4\%)}$

View cluster-by-cluster breakdown

Turbine Blade Failure: $P(C|\text{Cluster})=0.0882 \times P(\text{Cluster}|Q)=0.2760 = 0.0244$

4. Personnel issues-Task performance-Maintenance-Installation-Maintenance personnel - C

Calculation: $\sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0882 \times 0.2760) = 0.0244 \text{ (2.4\%)}$

View cluster-by-cluster breakdown

Turbine Blade Failure: $P(C|\text{Cluster})=0.0882 \times P(\text{Cluster}|Q)=0.2760 = 0.0244$

5. The fracture of the fan mid shaft due to a stress corrosion cracking initiated by local regions of corrosion pitting, which resulted from oil migration within the shaft and its subsequent degradation.

Calculation: $\sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0882 \times 0.2760) = 0.0244 \text{ (2.4\%)}$

View cluster-by-cluster breakdown

Turbine Blade Failure: $P(C|\text{Cluster})=0.0882 \times P(\text{Cluster}|Q)=0.2760 = 0.0244$

6. LEAKAGE OF THE NUMBER 7 FUEL NOZZLE, WHICH RESULTED IN AN IN FLIGHT ENGINE FIRE

Calculation: $\sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0435 \times 0.5355) = 0.0233 \text{ (2.3\%)}$

View cluster-by-cluster breakdown

View cluster-by-cluster breakdown

Engine Fire: $P(C|Cluster)=0.0435 \times P(Cluster|Q)=0.5355 = 0.0233$

7. THE TURBINE GUIDE VANE FAILURE

Calculation: $\sum P(Cause|Cluster) \times P(Cluster|Query) = (0.0435 \times 0.5355) = 0.0233$ (2.3%)

View cluster-by-cluster breakdown

Engine Fire: $P(C|Cluster)=0.0435 \times P(Cluster|Q)=0.5355 = 0.0233$

8. THE MALFUNCTION OF THE STARTER CONTROL VALVE

Calculation: $\sum P(Cause|Cluster) \times P(Cluster|Query) = (0.0435 \times 0.5355) = 0.0233$ (2.3%)

View cluster-by-cluster breakdown

Engine Fire: $P(C|Cluster)=0.0435 \times P(Cluster|Q)=0.5355 = 0.0233$

9. Contributing to the failure was a combination of manufacturing defects of the fuel nozzle adapter as

Calculation: $\sum P(Cause|Cluster) \times P(Cluster|Query) = (0.0435 \times 0.5355) = 0.0233$ (2.3%)

View cluster-by-cluster breakdown

Engine Fire: $P(C|Cluster)=0.0435 \times P(Cluster|Q)=0.5355 = 0.0233$

10. Contributing to the incident was a second defect of the fuel nozzle adapter assembly by the manufact

Calculation: $\sum P(Cause|Cluster) \times P(Cluster|Query) = (0.0435 \times 0.5355) = 0.0233$ (2.3%)

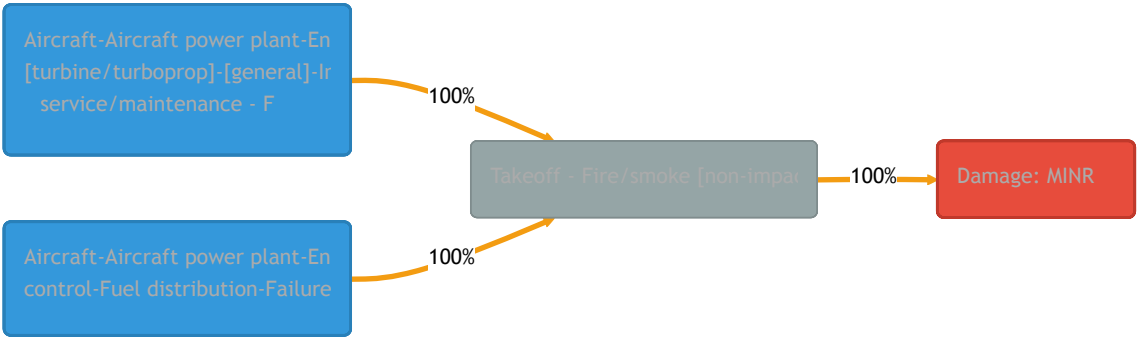
View cluster-by-cluster breakdown

Engine Fire: $P(C|Cluster)=0.0435 \times P(Cluster|Q)=0.5355 = 0.0233$

Final Diagnosis: Most Probable Causal Chain

What this shows: Based on all the calculations above, this diagram shows the precursors, events, and outcomes from the most relevant cluster. It combines all i show the most probable causal chain for your query.

Most relevant cluster: Engine Fire ($P(\text{Cluster}|\text{Query}) = 53.6\%$)



Top causes from the most relevant clusters:

Engine Fire ($P(\text{Cluster}|\text{Query}) = 53.6\%$):

- 2.3%: LEAKAGE OF THE NUMBER 7 FUEL NOZZLE, WHICH RESULTED IN AN IN FLIGHT ENGINE FIRE ($P(C|\text{Cluster}) = 4.3\%$)
- 2.3%: THE TURBINE GUIDE VANE FAILURE ($P(C|\text{Cluster}) = 4.3\%$)

Turbine Blade Failure ($P(\text{Cluster}|\text{Query}) = 27.6\%$):

- 2.4%: Personnel issues-Action/decision-Action-Incorrect action performance-Maintenance personnel - C ($P(C|\text{Cluster}) = 8.8\%$)
- 2.4%: Personnel issues-Task performance-Maintenance-Installation-Maintenance personnel - C ($P(C|\text{Cluster}) = 8.8\%$)

Evacuation Slide Malfunction ($P(\text{Cluster}|\text{Query}) = 6.3\%$):

- 3.1%: A passenger sustained a hairline fracture during an emergency evacuation of the airplane ($P(C|\text{Cluster}) = 50.0\%$)
- 3.1%: THE HIGH CYCLE FATIGUE FAILURE OF BLADES ON THE STAGE 1 HIGH PRESSURE COMPRESSOR, WHICH RESULTED IN BLADE FRACTURE AND SUBSEQUENT FIRE. ($P(C|\text{Cluster}) = 50.0\%$)

Similar Historical Incidents

These are the actual NTSB incident reports most similar to your query, ranked by cosine similarity score. Click to expand and see the full narrative, causes, and sequence of events.

Showing top 5 of 48 retrieved incidents. Click to expand and view details.

Incident: 20120221X03231 (Similarity: 70.90%)
Incident: 20161028X72837 (Similarity: 68.17%)
Incident: 20110808X94427 (Similarity: 68.16%)
Incident: 20111011X01145 (Similarity: 68.16%)
Incident: 20001211X12525 (Similarity: 68.12%)

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